$X_1 \sim N(\mu_1, \sigma_1)$

x_1

$X_2 \sim N(\mu_2, \sigma_2)$

x_2

$X \sim N(\mu_X, \sigma_X)$ $Y \sim N(\mu_Y, \sigma_Y)$

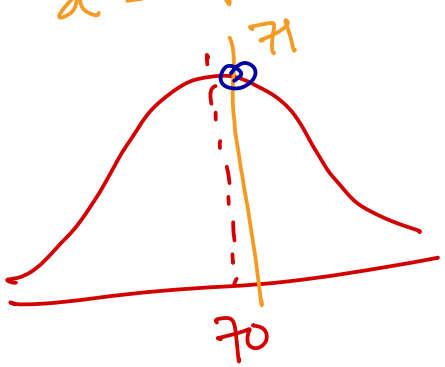
X = The height of male participant

Y = " " " female participant

$\mu_X = 70$ in

$\sigma_X = 3$ in

$x = 71$ in



P ~~dnorm~~ $(71, 70, 3)$

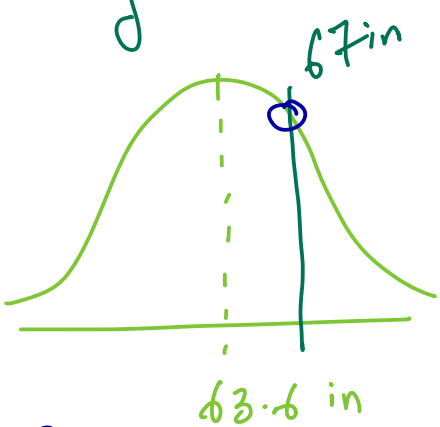
$= 0.126$

$= 0.63$

$\mu_Y = 63.6$ in

$\sigma_Y = 2.5$ in

$y = 67$ in



P ~~dnorm~~ $(67, 63.6, 2.5)$

$= 0.063$

$= 0.91$

Standardization: $z = \frac{x - \mu}{\sigma}$ → value of X; $\mu \to E(X)$; $\sigma \to \text{std}(X)$

if $X \sim N(\mu, \sigma)$

then $Z \sim N(0,1)$

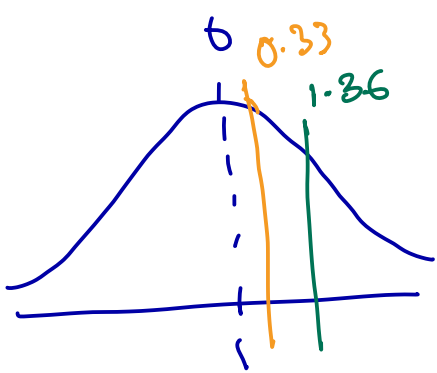
$$\boxed{z = \frac{x-\mu}{\sigma} \sim N(0,1)}$$

$$z_x = \frac{71-70}{3}$$

$$= 0.333$$

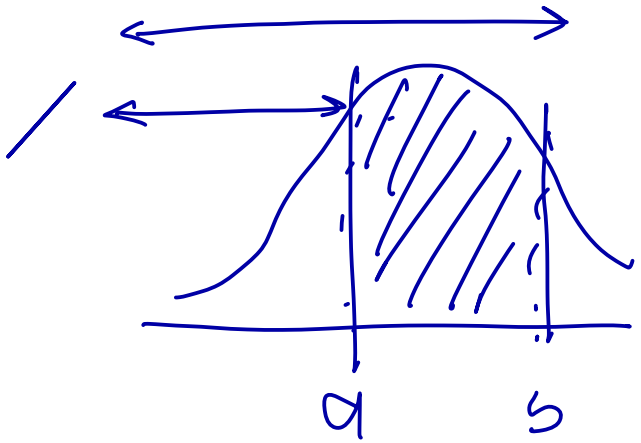
$$z_y = \frac{67-63.6}{2.5}$$

$$= 1.36$$



pnorm(0.33) = 0.6293

pnorm(1.36) = 0.91



$P(a < X \leq b)$

$$z = \frac{x - \mu_x}{\sigma_x}$$

$$z = \frac{y - \mu_y}{\sigma_y}$$

$$y = \sigma_y z + \mu_y$$

Central Limit Theorem (CLT)

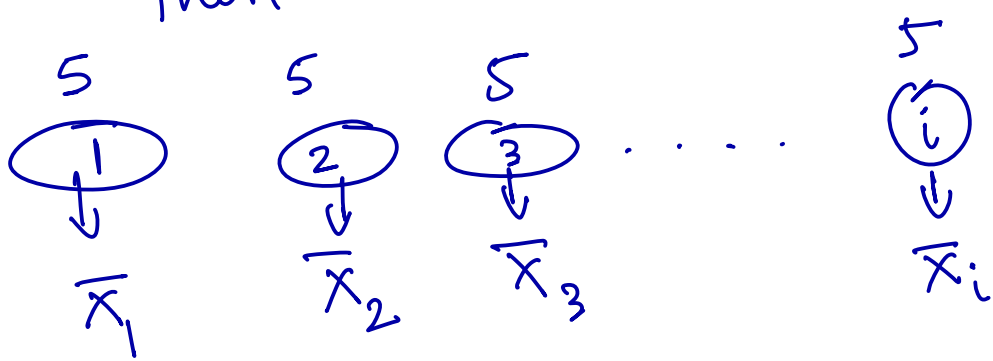
Any random variable

$$X \sim D$$

$$E(X) = \mu, \quad Std(X) = \sigma$$

Sample of $X \rightarrow$ size $= n$

then $$\bar{X} = \frac{1}{n} \Sigma X_i$$



$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$\frac{\sigma}{\sqrt{n}}$ = sampling error.

Condition:

If X is a non-normal dist

then $n \geq 30$.

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

If X is a normal dist

then $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

X is r.v with mean $= 24$ & $\sigma = 0.25$ & for a sample size of 300 what is the dist of $\bar{X}$?

$n \geq 30$

CLT $\Rightarrow \bar{X} \sim N\left(24, \frac{0.25}{\sqrt{300}}\right)$